

B3 Quadratic Functions

Name: _____

Class: _____

Date: _____

Time: **371 minutes**

Marks: **309 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

- (a) Express $2x^2 - 5x + k$ in the form $a(x - b)^2 + c$

(3)

- (b) Find the values of k for which the curve $y = 2x^2 - 5x + k$ does **not** intersect the line $y = 3$

(3)

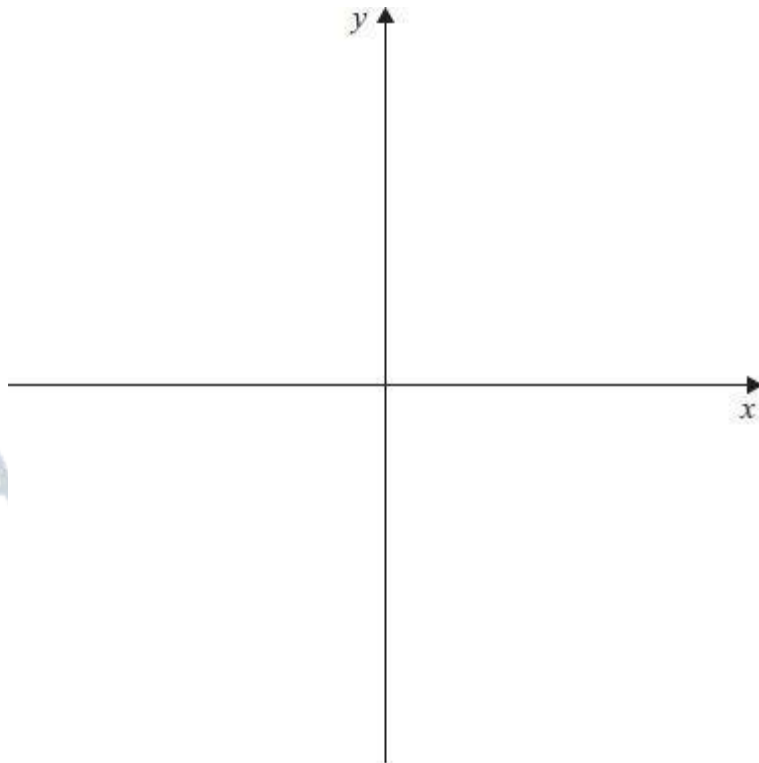
(Total 6 marks)

Q2.

A curve, C , has equation $y = x^2 - 6x + k$, where k is a constant.

The equation $x^2 - 6x + k = 0$ has two distinct positive roots.

- (a) Sketch C on the axes below.



(2)

- (b) Find the range of possible values for k .

Fully justify your answer.

(4)

(Total 6 marks)

Q3.

Find the coordinates, in terms of a , of the minimum point on the curve $y = x^2 - 5x + a$, where a is a constant.

Fully justify your answer.

(Total 3 marks)

Q4.

The quadratic equation $3x^2 + 4x + (2k - 1) = 0$ has real and distinct roots.

Find the possible values of the constant k
Fully justify your answer.

(Total 4 marks)

Q5.

Solve the equation

$$\sin\theta \tan\theta + 2\sin\theta = 3\cos\theta \quad \text{where } \cos\theta \neq 0$$

Give **all** values of θ to the nearest degree in the interval $0^\circ < \theta < 180^\circ$
Fully justify your answer.

(Total 5 marks)

Q6.

Prove that the function $f(x) = x^3 - 3x^2 + 15x - 1$ is an increasing function.

(Total 6 marks)

Q7.

Find the values of k for which the equation $(2k - 3)x^2 - kx + (k - 1) = 0$ has equal roots.

(Total 4 marks)

Q8.

A circle with centre C has equation $x^2 + y^2 + 8x - 12y = 12$

(a) Find the coordinates of C and the radius of the circle.

(3)

(b) The points P and Q lie on the circle.
The origin is the midpoint of the chord PQ .

Show that PQ has length $n\sqrt{3}$, where n is an integer.

(5)

(Total 8 marks)

Q9.

A curve, C , has equation $y = x^2 - 4x + k$, where k is a constant.

It crosses the x -axis at the points $(2 + \sqrt{5}, 0)$ and $(2 - \sqrt{5}, 0)$

- (a) Find the value of k .

(2)

- (b) Sketch the curve C , labelling any intersections with the axes.

(3)

(Total 5 marks)

Q10.

- (a) Three consecutive terms in an arithmetic sequence are $3e^{-p}$, 5 , $3e^p$

Find the possible values of p . Give your answers in an exact form.

(6)

- (b) Prove that there is no possible value of q for which $3e^{-q}$, 5 , $3e^q$ are consecutive terms of a geometric sequence.

(4)

(Total 10 marks)

Q11.

In this question use $g = 9.81 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

A ball is projected from the origin. After 2.5 seconds, the ball lands at the point with position vector $(40\mathbf{i} - 10\mathbf{j})$ metres.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively.

Assume that there are no resistance forces acting on the ball.

- (a) Find the speed of the ball when it is at a height of 3 metres above its initial position.

(6)

- (b) State the speed of the ball when it is at its maximum height.

(1)

- (c) Explain why the answer you found in part (b) may not be the actual speed of the ball when it is at its maximum height.

(1)

(Total 8 marks)

Q12.

- (a) (i) Express $x^2 - 6x + 11$ in the form $(x - p)^2 + q$.

(2)

- (ii) **Use the result from part (a)(i)** to show that the equation $x^2 - 6x + 11 = 0$ has no real solutions.

(2)

- (b) A curve has equation $y = x^2 - 6x + 11$.
- (i) Find the coordinates of the vertex of the curve. (2)
- (ii) Sketch the curve, indicating the value of y where the curve crosses the y -axis. (3)
- (iii) Describe the geometrical transformation that maps the curve with equation $y = x^2 - 6x + 11$ onto the curve with equation $y = x^2$. (3)
- (Total 12 marks)**

Q13.

A circle with centre $C(-3, 2)$ has equation

$$x^2 + y^2 + 6x - 4y = 12$$

- (a) Find the y -coordinates of the points where the circle crosses the y -axis. (3)
- (b) Find the radius of the circle. (3)
- (c) The point $P(2, 5)$ lies outside the circle.
- (i) Find the length of CP , giving your answer in the form $\sqrt{n}$, where n is an integer. (2)
- (ii) The point Q lies on the circle so that PQ is a tangent to the circle. Find the length of PQ . (2)
- (Total 10 marks)**

Q14.

A curve has equation $y = 2x^2 - x - 1$ and a line has equation $y = k(2x - 3)$, where k is a constant.

- (a) Show that the x -coordinate of any point of intersection of the curve and the line satisfies the equation
- $$2x^2 - (2k + 1)x + 3k - 1 = 0$$
- (1)
- (b) The curve and the line intersect at two distinct points.
- (i) Show that $4k^2 - 20k + 9 > 0$. (3)

- (ii) Find the possible values of k .

(4)

(Total 8 marks)

Q15.

- (a) (i) Express $2x^2 + 6x + 5$ in the form $2(x + p)^2 + q$, where p and q are rational numbers.

(2)

- (ii) Hence write down the minimum value of $2x^2 + 6x + 5$.

(1)

- (b) The point A has coordinates $(-3, 5)$ and the point B has coordinates $(x, 3x + 9)$.

- (i) Show that $AB^2 = 5(2x^2 + 6x + 5)$.

(3)

- (ii) Use your result from part (a)(ii) to find the minimum value of the length AB as x varies, giving your answer in the form $\frac{1}{2}\sqrt{n}$, where n is an integer.

(2)

(Total 8 marks)

Q16.

The quadratic equation

$$(2k - 7)x^2 - (k - 2)x + (k - 3) = 0$$

has real roots.

- (a) Show that $7k^2 - 48k + 80 \leq 0$.

(4)

- (b) Find the possible values of k .

(4)

(Total 8 marks)

Q17.

- (a) Given that $\log_a b = c$, express b in terms of a and c .

(1)

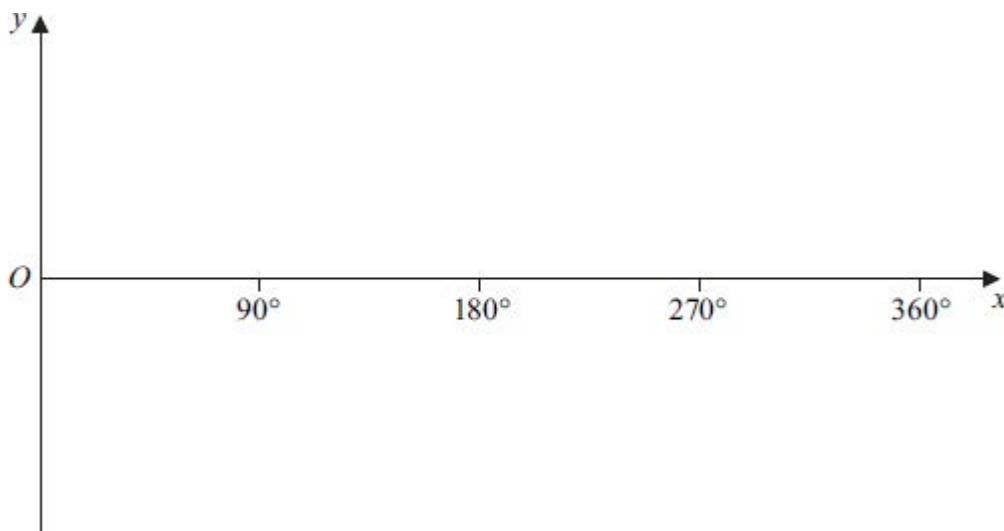
- (b) By forming a quadratic equation, show that there is only one value of x which satisfies the equation $2 \log_2(x + 7) - \log_2(x + 5) = 3$.

(6)

(Total 7 marks)

Q18.

- (a) (i) On the axes given below, sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$.



(3)

- (ii) Solve the equation $\tan x = -1$, giving all values of x in the interval $0^\circ \leq x \leq 360^\circ$.

(2)

- (b) (i) Given that $6 \tan \theta \sin \theta = 5$, show that $6 \cos^2 \theta + 5 \cos \theta - 6 = 0$.

(3)

- (ii) **Hence** solve the equation $6 \tan 3x \sin 3x = 5$, giving all values of x to the nearest degree in the interval $0^\circ \leq x \leq 180^\circ$.

(6)

(Total 14 marks)

Q19.

By forming and solving a quadratic equation, solve the equation

$$8 \sec x - 2 \sec^2 x = \tan^2 x - 2$$

in the interval $0 < x < 2\pi$, giving the values of x in radians to three significant figures.

(Total 7 marks)

Q20.

- (a) Factorise $x^2 - 4x - 12$.

(1)

- (b) Sketch the graph with equation $y = x^2 - 4x - 12$, stating the values where the curve crosses the coordinate axes.

(4)

- (c) (i) Express $x^2 - 4x - 12$ in the form $(x - p)^2 - q$, where p and q are positive integers.

(2)

(ii) Hence find the minimum value of $x^2 - 4x - 12$.

(1)

(d) The curve with equation $y = x^2 - 4x - 12$ is translated by the vector $\begin{bmatrix} -3 \\ 2 \end{bmatrix}$.

Find an equation of the new curve. You need not simplify your answer.

(2)

(Total 10 marks)

Q21.

A circle with centre C has equation $x^2 + y^2 + 14x - 10y + 49 = 0$.

(a) Express this equation in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

(b) Write down:

(i) the coordinates of C ;

(3)

(ii) the radius of the circle.

(2)

(c) Sketch the circle.

(2)

(d) A line has equation $y = kx + 6$, where k is a constant.

(i) Show that the x -coordinates of any points of intersection of the line and the circle satisfy the equation $(k^2 + 1)x^2 + 2(k + 7)x + 25 = 0$.

(2)

(ii) The equation $(k^2 + 1)x^2 + 2(k + 7)x + 25 = 0$ has equal roots. Show that

$$12k^2 - 7k - 12 = 0$$

(3)

(iii) Hence find the values of k for which the line is a tangent to the circle.

(2)

(Total 14 marks)

Q22.

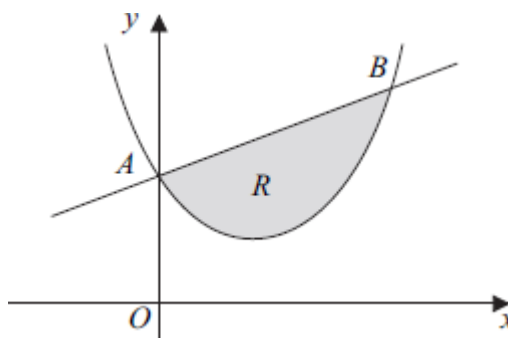
(a) (i) Express $x^2 - 3x + 5$ in the form $(x - p)^2 + q$.

(2)

- (ii) Hence write down the equation of the line of symmetry of the curve with equation $y = x^2 - 3x + 5$.

(1)

- (b) The curve C with equation $y = x^2 - 3x + 5$ and the straight line $y = x + 5$ intersect at the point $A(0, 5)$ and at the point B , as shown in the diagram below.



- (i) Find the coordinates of the point B .

(3)

- (ii) Find $\int (x^2 - 3x + 5) dx$.

(3)

- (iii) Find the area of the shaded region R bounded by the curve C and the line segment AB .

(4)

(Total 13 marks)

Q23.

- (a) Sketch the curve with equation $y = 7^x$, indicating the coordinates of any point where the curve intersects the coordinate axes.

(2)

- (b) The curve C_1 has equation $y = 7^x$.
The curve C_2 has equation $y = 7^{2x} - 12$.

- (i) By forming and solving a quadratic equation, prove that the curves C_1 and C_2 intersect at exactly one point. State the y -coordinate of this point.

(4)

- (ii) Use logarithms to find the x -coordinate of the point of intersection of C_1 and C_2 , giving your answer to three significant figures.

(2)

(Total 8 marks)

Q24.

The curve with equation $y = 13 + 18x + 3x^2 - 4x^3$ passes through the point P where $x = -1$.

- (a) Find $\frac{dy}{dx}$ (3)
- (b) Show that the point P is a stationary point of the curve and find the other value of x where the curve has a stationary point (3)
- (c) (i) Find the value of $\frac{d^2y}{dx^2}$ at the point P . (3)
- (ii) Hence, or otherwise, determine whether P is a maximum point or a minimum point. (1)
- (Total 10 marks)**

Q25.

The line AB has equation $3x + 2y = 7$. The point C has coordinates $(2, -7)$.

- (a) (i) Find the gradient of AB . (2)
- (ii) The line which passes through C and which is parallel to AB crosses the y -axis at the point D . Find the y -coordinate of D . (3)
- (b) The line with equation $y = 1 - 4x$ intersects the line AB at the point A . Find the coordinates of A . (3)
- (c) The point E has coordinates $(5, k)$. Given that CE has length 5, find the two possible values of the constant k . (3)
- (Total 11 marks)**

Q26.

- (a) (i) Sketch the curve with equation $y = x(x - 2)^2$. (3)
- (ii) Show that the equation $x(x - 2)^2 = 3$ can be expressed as
- $$x^3 - 4x^2 + 4x - 3 = 0$$
- (1)

- (b) The polynomial $p(x)$ is given by $p(x) = x^3 - 4x^2 + 4x - 3$.
- (i) Use the Factor Theorem to show that $x - 3$ is a factor of $p(x)$. (2)
- (ii) Express $p(x)$ in the form $(x - 3)(x^2 + bx + c)$, where b and c are integers. (2)
- (c) Hence show that the equation $x(x - 2)^2 = 3$ has only one real root and state the value of this root. (3)
- (Total 11 marks)**

Q27.

- (a) (i) Express $4 - 10x - x^2$ in the form $p - (x + q)^2$ (2)
- (ii) Hence write down the equation of the line of symmetry of the curve with equation $y = 4 - 10x - x^2$. (1)
- (b) The curve C has equation $y = 4 - 10x - x^2$ and the line L has equation $y = k(4x - 13)$, where k is a constant.
- (i) Show that the x -coordinates of any points of intersection of the curve C with the line L satisfy the equation
- $$x^2 + 2(2k + 5)x - (13k + 4) = 0$$
- (1)
- (ii) Given that the curve C and the line L intersect in two distinct points, show that
- $$4k^2 + 33k + 29 > 0$$
- (3)
- (iii) Solve the inequality $4k^2 + 33k + 29 > 0$. (4)
- (Total 11 marks)**

Q28.

- (a) (i) Given that $\tan 2x + \tan x = 0$, show that $\tan x = 0$ or $\tan^2 x = 3$. (3)
- (ii) Hence find all solutions of $\tan 2x + \tan x = 0$ in the interval $0^\circ < x < 180^\circ$. (1)
- (b) (i) Given that $\cos x \neq 0$, show that the equation

$$\sin 2x = \cos x \cos 2x$$

can be written in the form

$$2 \sin^2 x + 2 \sin x - 1 = 0$$

(3)

- (ii) Show that all solutions of the equation $2 \sin^2 x + 2 \sin x - 1 = 0$ are given by

$$\sin x = \frac{\sqrt{3} - 1}{p}, \text{ where } p \text{ is an integer.}$$

(3)

(Total 10 marks)

Q29.

- (a) Express $x^2 + 5x + 7$ in the form $(x + p)^2 + q$, where p and q are rational numbers.

(3)

- (b) A curve has equation $y = x^2 + 5x + 7$.

- (i) Find the coordinates of the vertex of the curve.

(2)

- (ii) State the equation of the line of symmetry of the curve.

(1)

- (iii) Sketch the curve, stating the value of the intercept on the y -axis.

(3)

- (c) Describe the geometrical transformation that maps the graph of $y = x^2$ onto the graph of $y = x^2 + 5x + 7$.

(3)

(Total 12 marks)

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Q30.

The polynomial $p(x)$ is given by $p(x) = x^3 - 2x^2 + 3$.

- (a) Use the Factor Theorem to show that $x + 1$ is a factor of $p(x)$.

(2)

- (b) (i) Express $p(x) = x^3 - 2x^2 + 3$ in the form $(x + 1)(x^2 + bx + c)$, where b and c are integers.

(2)

- (ii) Hence show that the equation $p(x) = 0$ has exactly one real root.

(2)

(Total 6 marks)

Q31.

A circle has centre C (3, -8) and radius 10.

- (a) Express the equation of the circle in the form

$$(x - a)^2 + (y - b)^2 = k \quad (2)$$

- (b) Find the x -coordinates of the points where the circle crosses the x -axis. (3)

- (c) The tangent to the circle at the point A has gradient $\frac{5}{2}$. Find an equation of the line CA , giving your answer in the form $rx + sy + t = 0$, where r , s and t are integers. (3)

- (d) The line with equation $y = 2x + 1$ intersects the circle.

- (i) Show that the x -coordinates of the points of intersection satisfy the equation

$$x^2 + 6x - 2 = 0 \quad (3)$$

- (ii) Hence show that the x -coordinates of the points of intersection are of the form $m \pm \sqrt{n}$, where m and n are integers. (2)
(Total 13 marks)

Q32.

- (a) Given that $3 \ln x = 4$, find the exact value of x . (1)

- (b) By forming a quadratic equation in $\ln x$, solve $3 \ln x + \frac{20}{\ln x} = 19$, giving your answers for x in an exact form. (5)
(Total 6 marks)

Q33.

- (a) On separate diagrams:

- (i) sketch the curve with equation $y = |3x + 3|$; (2)

- (ii) sketch the curve with equation $y = |x^2 - 1|$. (3)

- (b) (i) Solve the equation $|3x + 3| = |x^2 - 1|$. (5)
- (ii) Hence solve the inequality $|3x + 3| < |x^2 - 1|$. (2)
- (Total 12 marks)

Q34.

- (a) Express $(x - 5)(x - 3) + 2$ in the form $(x - p)^2 + q$, where p and q are integers. (3)
- (b) (i) Sketch the graph of $y = (x - 5)(x - 3) + 2$, stating the coordinates of the minimum point and the point where the graph crosses the y -axis. (3)
- (ii) Write down an equation of the tangent to the graph of $y = (x - 5)(x - 3) + 2$ at its vertex. (2)
- (c) Describe the geometrical transformation that maps the graph of $y = x^2$ onto the graph of $y = (x - 5)(x - 3) + 2$. (3)
- (Total 11 marks)

Q35.

- (a) (i) Express $2x^2 - 20x + 53$ in the form $2(x - p)^2 + q$, where p and q are integers. (2)
- (ii) Use your result from part (a)(i) to explain why the equation $2x^2 - 20x + 53 = 0$ has no real roots. (2)
- (b) The quadratic equation $(2k - 1)x^2 + (k + 1)x + k = 0$ has real roots.
- (i) Show that $7k^2 - 6k - 1 \leq 0$. (4)
- (ii) Hence find the possible values of k . (4)
- (Total 12 marks)